IMPERIAL
BUSINESS SCHOOL

Introduction to Data Science
Group assignment

Measuring progress towards the UN Sustainable Development Goal 8: Decent work and economic growth

The United Nations Sustainable Development Goals (SDGs) are targets for global development that were adopted in 2015. All countries have agreed to work towards achieving them by 2030.

In this group assignment, you will assess different continents' progress towards meeting the UN Sustainable Development Goal 8: Decent Work and Economic Growth. To promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.

You are given the following data sets in csv files:

1. **continents-according-to-our-world-in-data.csv**
    - This data set classifies countries into seven different continents: Asia, Africa, North America, South America, Antarctica, Europe, or Oceania.

2. **gdp-per-capita-worldbank.csv**
    - This data set contains the GDP per capita, reported in 2017 USD$, using the purchasing power parity (PPP) exchange rate for all countries in the world, going back to 1990 (if the data is available).

3. **youth-not-in-education-employment-training.csv**
    - This data set contains the share of youth not in education, employment or training, reported as a % of the total youth population for all countries in the world, going back to 1976 (if the data is available). Youth is considered to be people of ages 15 to 24.

Assignment brief

- Perform exploratory data analysis to assess how 6 continents in the world (all continents except Antarctica) are faring with respect to the following specific targets set out by the UN:

1. Sustain per capita economic growth in accordance with national circumstances and, in particular, at least 7 per cent gross domestic product growth per annum in the least

developed countries.

2. By 2020, substantially reduce the proportion of youth not in employment, education or training.

- Locate, in collaboration with your group, an additional CSV file which can add to your analysis. This CSV file should add further information which you can incorporate into your dataset. The provenance of the CSV file should be assessed and discussed.

Assignment format and handin

- Your analysis should draw together information from the three CSV files provided as well as from the supplemental CSV file located by your team.
- Present your findings in a presentation recording of no more than 15 minutes in length (one video per group). Not all group members need to present in the video.
- If you use external help or resources, you must cite them and include the full bibliography in a separate References section using the Harvard referencing style. You should include any references as footnotes or a separate slide at the end of your report (this does not count towards the 5 page limit).
- Large language models (LLMs) produce vague, inconsistent and often incorrect output. The use of LLMs is not prohibited, however work which is vague, inconsistent or incorrect will be marked accordingly.
- Your work will be marked in accordance with the Imperial undergraduate marking guidelines.
- Submit your video as an .mp4 file via the Hub.
- Video files can be large, however they can usually be converted to reduce their size. If your video file is too large to upload on the Hub, use a video conversion program such as Handbrake to reduce its size; care should be taken that all text is readable. If you have problems with this, ask for assistance.
- Video files must not be submitted as file sharing (Onedrive, Dropbox) links; the file itself must be uploaded. If you have problems with submission, contact the Programme Team.

Guidelines

- Consider what each target means carefully. There may not be one “correct” measure, so you may want to explore more than one to assess each target.
- You may wish to compare across the continents, as well as assess their individual progress across time. You may also wish to further divide up a continent (e.g., into more developed vs. less developed countries).
- Explain what data processing or analysis you did, and why.
- Try to complete all analysis early and focus on the presentation, report and cleaning the codes, as putting analysis together takes time.
- Work steadily. You would need to devote 2-4 hours each week leading up to the deadline on the assignment. Do not leave this assignment to the last minute.
- Try to use the R functions you learnt in class first, before trying to find more advanced functions. As we will continue to cover more R in class, revisit the assignment with the new knowledge each week.

- Structure your presentation to have the following sections: 1. Introduction 2. Methods 3. Results 4. Discussion and 5. Conclusion.

Mark Breakdown:

	Details	**Marks**
Video presentation	- Video presentation is clear and precise (10) - Analyses are clearly explained (10) - Design decisions are justified (10)	**30**
Data analysis: Target 1	- Data sets are analysed in sufficient detail to assess the targets (5) - Any data processing/analysis are well-justified and thought-through (10) - Analysis is expressed in clear and meaningful ways (10)	**20**
Data analysis: Target 2	- Data sets are analysed in sufficient detail to assess the targets (5) - Any data processing/analysis are well-justified and thought-through (10) - Analysis is expressed in clear and meaningful ways (10)	**20**
Code	- Code is presented and explained clearly and concisely (10) - Analysis pipeline decisions are explained and justified (10)	**20**
References	- References should be clearly explained in the presentation (5) - Citations should be provided either as footnotes or on a references slide at the end (5)	**10**
Total		**100**